

Speaker notes — Semantic Dial Compass

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Target length: ≈ 12 minutes + Q&A. Each **beat** box is one slide-equivalent. The phrasing is conversational — do not read verbatim; use the bullets as landmarks. Time stamps are cumulative.

0. Hook (~30s)

Say: “Billion-parameter language models are black boxes. Today I’ll show you a single 1-dimensional direction — one axis — inside one attention head, that cleanly encodes gender. Turn that one knob and bias on WinoGender drops to zero. Fluency stays put. Same story reproduces in four unrelated model families.”

Beat: hold a second after “one axis.” That is the claim.

1. What we set out to do (~1:00)

Three claims we needed to prove:

1. such a *dial* exists inside the model,
2. it is **causal** — the model really uses it,
3. we can **turn it up or down** to change behaviour.

Anything less than all three is not interpretability, it is just staring at weights.

2. Where to look: W_{OV} and the SVD (~2:00)

One sentence of mechanism: each attention head writes back through a matrix $W_{OV} = W_V W_O$. SVD gives an ordered basis of *what this head can write*. The top rows of $V^\top$ are the head’s strongest write directions — we call these **candidate compass axes**.

Slide cue: show the $W_{OV} = U \Sigma V^\top$ line and circle rows of $V^\top$.

Pitfall to avoid on stage: don’t get pulled into a full transformer tutorial. Say “it’s a write direction” and move on.

3. The scan — “is this plane a compass?” (~3:30)

Setup: for each (layer, head, 2-D plane of top SVs) we inject a rotating 2-D signal and watch one readout — the logit gap between the pronouns “he” and “she” on three neutral prompts. No WinoGender, no BLS labels. Just a probe.

Two tests:

- **Amplitude linearity:** doubling input $\Rightarrow$ doubling output ($R^2 > 0.95$).
- **Phase stability:** output angle stays put ($|\Delta\varphi| < 10^\circ$).

Key number to land: Compasses are rare. Pass rates: GPT-2 1.9%, Phi-3 1.0%, Gemma 1.2%, Llama-3.2-3B 0.2%. “Most heads are noise; a tiny fraction carry clean signal.”

4. Null tests — are we fooling ourselves? (~5:00)

The reviewer’s worry: 2% is not big. Could any random 2-D plane look compass-like?

Two nulls:

- **Loose:** random planes from the whole output space. Compass **wins** on GPT-2, Phi-3, Gemma.
- **Strict:** random planes from only the head’s top-4 SVs. Compass **ties** — the top-4 subspace is densely compass-like.

Honest framing (memorise this sentence): “Our claim is not ‘we found *the* gender axis’ — it is *gender is concentrated in a low-dim subspace of one attention head*.” The tie is a feature, not a failure: it tells us *where inside the head gender lives*.

5. What does the axis encode? (~6:30)

Decode the axis through the unembedding W_U and read the top tokens.

- GPT-2 L10 H9 (SV0): *he, him, his, man* vs *she, her, woman*.
- Phi-3 L28 H1 (SV0): *his, lui,* vs *her, elle, ella* — multilingual.
- Gemma L21 H4 (SV0): *he, , él* vs *she, .*
- Llama-3.2-3B L26 H14 (**SV2, not SV0!**): *she, her, ,* vs *his, , jago*.

Llama quirk — call it out explicitly: SV0 and SV1 on L26 H14 decode to register/punctuation noise. Gender sits at SV2. Same causal story, slightly different location. This is actually evidence *for* the mechanism (it’s language-agnostic semantic gender, not a top-SV artifact).

6. Is it causal? the α -sweep (~8:00)

The key test. Projection ablation at the compass head’s layer: $r' = r - \alpha \langle r, u \rangle u$ for $\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2\}$. $\alpha=1$ zeros the axis; $\alpha=2$ over-shoots.

Evaluate on WinoGender with two metrics (**stereo_corr** against BLS, and **stereo_delta**).

Headline to land slowly:

1. Bias drops **linearly** with α .
2. Hits **zero near** $\alpha=1.5$.
3. **Reverses sign** at $\alpha=2$.
4. Perplexity barely moves (≤ 1 PPL point; +0.035 PPL on Llama).

“That is the shape of a real dial, not a correlation.”

7. Baseline comparison vs ActAdd (~9:00)

Framing: ActAdd (Turner et al. 2023) is the standard direction-based steering baseline — it takes mean(he-prompts) – mean(she-prompts) at a whole layer.

Result: compass at $\alpha=1.5$ **beats** ActAdd at $\alpha=1.0$ on all four models, at 4–20× lower perplexity cost.

The sentence: “compass is one targeted axis inside one head; ActAdd is a diffuse whole-layer direction. Targeted beats diffuse.”

If pressed on Llama specifically: compass moves **stereo_corr** from 0.73 to -0.18 ; ActAdd barely budges it ($0.73 \rightarrow 0.72$).

8. Cherry-pick defense + head specificity (~10:00)

Anticipate the objection: “you picked the best head after seeing results.”

Counter: apply the same $\alpha=1$ edit to *other* passing compass heads in each model. They **do not debias**. GPT-2 L9 H7 / L11 H1 move `stereo_corr` by < 0.02 . Phi-3 same story.

What those other heads encode (the taxonomy slide): number (3pl vs 3sg), person (1/2 vs 3), demonstratives, kinship. *They pass the compass scan but encode different pronoun features.*

Payoff sentence: “This is a **dissociation**. The primary head is specifically the occupational-stereotype readout. The finding is sharper, not weaker.”

9. Cross-benchmark honesty — StereoSet (~11:00)

Do not hide this. We ran StereoSet (sentence-level) as a second benchmark.

Result: WinoGender replicates on all 4; StereoSet replicates on GPT-2 only. On Phi-3/Gemma/Llama the compass acts at the pronoun slot but does not propagate into whole-sentence log-probs.

The framing: “This is a **boundary of the finding**. Compass is a position-local, gender-specific direction — not a universal bias knob. Domain specificity is also why race/profession/religion edits don’t transfer from the gender axis.”

Slide cue: show the StereoSet-domains figure; point at race and religion bars being flat.

10. Close (~12:00)

Say: “Four models. One head each. One 1-D axis inside that head. Zero it, bias vanishes. Over-shoot it, bias reverses. Fluency intact. That is a *surgical*, causal, human-meaningful dial — and we know what it encodes in four languages.”

Optional tag: “Interpretability doesn’t have to be post-hoc storytelling. Sometimes it’s finding the actual knob.”

Do not close with ‘thanks’ — close with the claim, then pause for questions.

Anticipated Q&A

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Q: Why only gender? Is this a universal bias knob?

A: No — and we *prove* that in the StereoSet-domains plot. The gender compass moves gender SS but barely touches race/profession/religion. That is evidence the axis is *specific*, which is a stronger claim than a generic debiaser.

Q: Isn’t the strict null tie a problem?

A: No, it is informative. It says gender lives in a small 2-D patch of directions near the top SVs, not on one magic axis. Our chosen axis is one clean member of that patch; the loose-null win is what proves the head carries gender.

Q: You tuned on WinoGender; isn’t $\alpha=1.5$ overfit?

A: The compass direction is built from **3 neutral prompts** with no WinoGender overlap. α is a 1-D scalar swept on an equal 6-point grid; no per-model tuning. The same grid yields the same zero-crossing across all four models.

Q: Why does Llama put gender at SV2 instead of SV0?

A: Llama-3.2-3B has a heavier-tailed W_{OV} spectrum for this head — SV0 and SV1 capture register and punctuation structure, SV2 captures gender. The mechanism is the same; the ordering inside the head differs. It’s a nice lesson: don’t assume SV0 is always where semantics lives.

Q: How does this compare to probing / INLP?

A: Probing finds *a* direction in the residual that predicts the label; it does not check the model *uses* it. Our causal α -sweep is the difference. We also ran INLP and SentenceDebias as baselines — see the per-domain CrowS-Pairs tables.

Q: Compute cost?

A: No retraining. Scan + causal sweep + decode runs in under an hour on a single A100 per model. All hooks are TransformerLens forward hooks.

Q: What doesn't work?

A: (a) StereoSet on Phi-3/Gemma/Llama — compass is position-local, SS is sentence-level. (b) Gemma religion — all 4 top heads fail the SNR gate, fallback retained; the routed ensemble cannot fix a domain the model does not localise. Both are in the paper as honest limits.

Cue-card reminders:

- Read the **four pass-rates** slowly (1.9 / 1.0 / 1.2 / 0.2 %).
- Say “*one 1-D axis*” out loud at least three times.
- Land “*reverses sign at $\alpha=2$* ” — this is the “aha.”
- **Never** say “the gender neuron.” Say “gender direction.”
- If a question lands cleanly, answer in two sentences and stop.